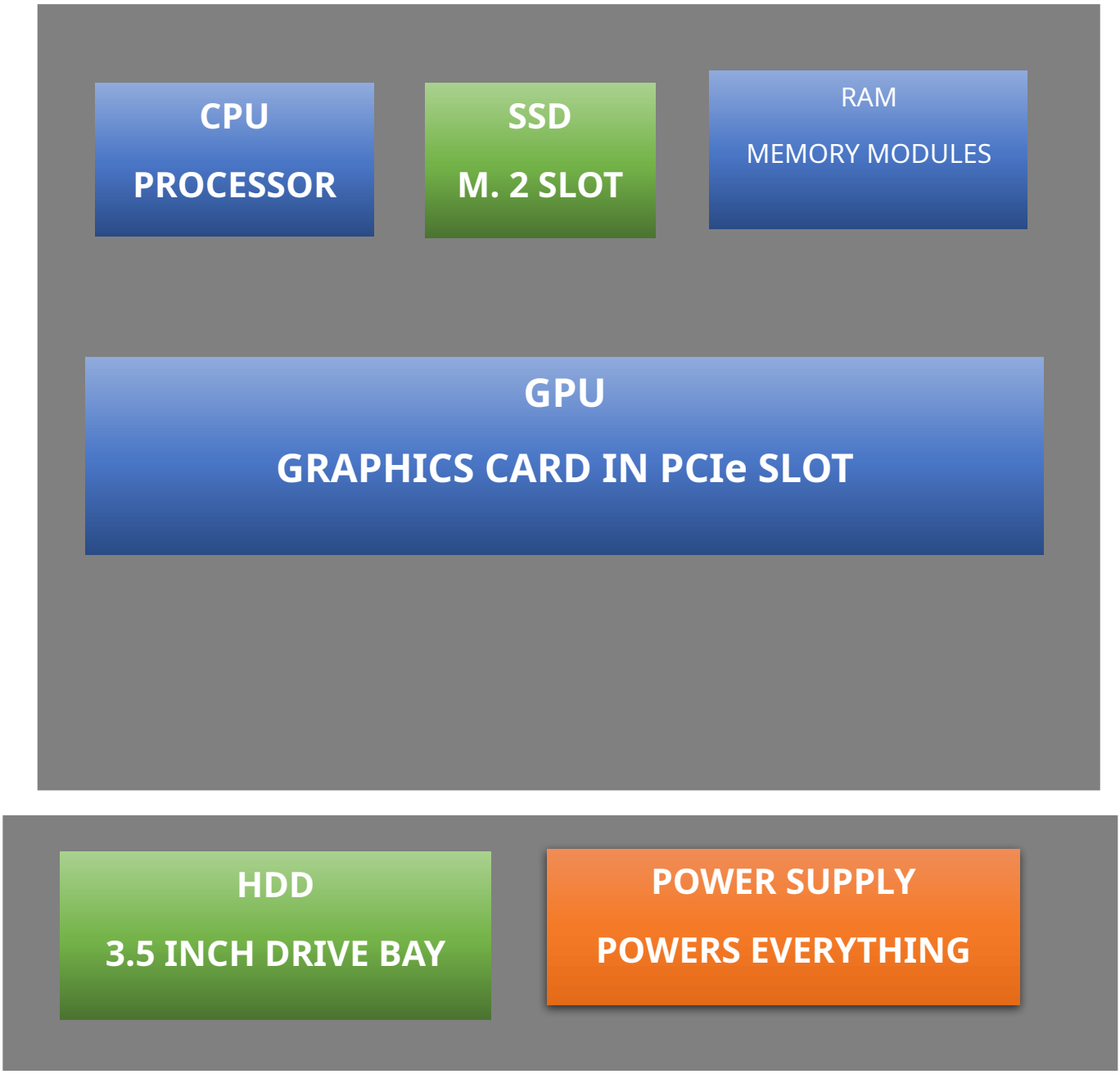
HARDWARE 1.1

DAY : 1

Q.1 Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.

ANS.1:-



**Q.2 Create a comparison table showing the main differences between HDD and SSD storage in terms of speed , durability , and typical use cases.

Hint: Think about which one is faster, Which Is more shock resistant, and where you would usually find each type (laptop, gaming PC, etc.).**

ANS:-2

	HDD	SDD
SPEED	**Slower (~80-160MB/s); Slow boot and load times**	**Much faster (~500 MB/s for SATA, 3000+ MB/s for NVMe); near instant boot**
DURABILITY	**Spinning platters and moving read/write heads make it vulnerable to drops and shocks**	**No moving parts, so it's far more shock-resistant**
TYPICAL USE	**Bluck storage, backups, NAS, older or budget desktops**	**Laptops, gaming PCs, workstations phones, and OS/boot drives**

Q.3 Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

ANS:-3

- **CPU: It runs the game logic: player movement, bullet physics, AI, and network data from otherplayers. A weak CPU causes stutters when many players are in the same area.**
- **RAM: It holds the map, textures, and game data the game needs right now, so they load quickly.Too little RAM makes the game lag or crash, especially with other apps open.**
- **GPU: It draws every frame you see: the terrain, buildings, characters, shadows, and effects. Astronger GPU gives higher FPS and better graphics settings.**

Q.4 Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

ANS:-4

- **GPU (first priority): FIFA is graphics-heavy, with detailed players, stadiums, and lighting. A better GPU gives smoother frame rates at higher settings.**
- **RAM: Go to at least 16 GB, since 8 GB can cause stutters with background apps running.**
- **SSD: Move the game onto an SSD for much faster loading screens and menus. An NVMe drive is best.**
- **CPU: Upgrade only if it is old, since it can bottleneck a new GPU and cause dips during busy matches.**
- **PSU: A new GPU may need more power, so check the wattage to keep the system stable.**

- **Motherboard: Replace it only if the new CPU or RAM is not compatible with your current one.**